

Joint Variation

Algebra Worksheet · Grade 9–11

Name: _____

Date: _____

Learning Objectives

- Identify joint variation relationships and write the general equation $y = kxz$
- Find the constant of variation k using given values of the variables
- Solve real-world and multi-step problems involving joint variation

Problems

1. y varies jointly as x and z . Write the general equation that models this relationship.

$$y = kxz$$

2. y varies jointly as x and z . If $y = 45$ when $x = 18$ and $z = 10$, find the constant of variation k .

$$y = kxz$$

3. y varies jointly as x and z . Given $y = 45$ when $x = 18$ and $z = 10$, find the equation of variation.

$$y = kxz$$

4. Using the equation of variation found in Problem 3, find y when $x = 20$ and $z = 30$.

$$y = \frac{1}{4}xz$$

5. The area A of a rectangle varies jointly as its length l and width w . If the area is 84 square units when the length is 12 units and the width is 7 units, find the constant of variation and write the equation of variation.

$$A = k/lw$$

6. The force F on an object varies jointly as its mass m and acceleration a . If $F = 200$ Newtons when $m = 25$ kg and $a = 8$ m/s squared, find the equation of variation and then find F when $m = 40$ kg and $a = 5$ m/s



squared.

$$F = kma$$

7. y varies jointly as x squared and z . If $y = 72$ when $x = 3$ and $z = 8$, find the constant of variation k , write the equation of variation, and then find y when $x = 5$ and $z = 4$.

$$y = kx^2z$$

8. The cost C of a plane ticket varies jointly as the number of layovers n and the distance d of the trip in miles. If a ticket costs \$480 for a trip with 2 layovers over a distance of 800 miles, find the equation of variation. Then find the cost for a trip with 3 layovers over a distance of 1200 miles.

$$C = knd$$

9. w varies jointly as x , y , and z . If $w = 210$ when $x = 3$, $y = 5$, and $z = 14$, find the constant of variation k and the equation of variation. Then find w when $x = 6$, $y = 4$, and $z = 10$.

$$w = kxyz$$

10. The volume V of a cone varies jointly as its height h and the square of its base radius r . A cone with height 9 cm and base radius 4 cm has a volume of 48π cubic centimeters. Find the constant of variation k and write the equation of variation. Then find the volume of a cone with height 12 cm and base radius 5 cm. Leave your answer in terms of π .

$$V = khr^2$$



Joint Variation — Answer Key

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Answer Key

1. Answer: $y = kxz$, where k is the constant of variation

- Joint variation means y is proportional to the product of two or more variables.
- The general formula is $y = kxz$, where k is the constant of variation.

2. Answer: $k = 1/4$

- Start with the formula: $y = kxz$.
- Substitute $y = 45$, $x = 18$, $z = 10$: $45 = k(18)(10)$.
- Simplify: $45 = 180k$.
- Divide both sides by 180: $k = 45/180 = 1/4$.

3. Answer: $y = (1/4)xz$

- Find k first: $45 = k(18)(10) \rightarrow 45 = 180k \rightarrow k = 1/4$.
- Substitute k back into $y = kxz$.
- The equation of variation is $y = (1/4)xz$.

4. Answer: $y = 150$

- Use the equation $y = (1/4)xz$.
- Substitute $x = 20$ and $z = 30$: $y = (1/4)(20)(30)$.
- Calculate: $y = (1/4)(600) = 150$.

5. Answer: $k = 1$, $A = lw$

- Write the joint variation equation: $A = klw$.
- Substitute $A = 84$, $l = 12$, $w = 7$: $84 = k(12)(7)$.
- Simplify: $84 = 84k$.
- Divide: $k = 1$, so the equation is $A = lw$.

6. Answer: $k = 1$, $F = ma$; $F = 200$ Newtons

- Write the joint variation equation: $F = kma$.
- Substitute $F = 200$, $m = 25$, $a = 8$: $200 = k(25)(8) = 200k \rightarrow k = 1$.
- The equation is $F = ma$.
- Find F when $m = 40$, $a = 5$: $F = (40)(5) = 200$ Newtons.

7. Answer: $k = 1$; $y = x^2z$; $y = 100$

- Write the equation: $y = kx^2z$.
- Substitute $y = 72$, $x = 3$, $z = 8$: $72 = k(9)(8) = 72k \rightarrow k = 1$.
- The equation is $y = x^2z$.
- Find y when $x = 5$, $z = 4$: $y = (25)(4) = 100$.

8. Answer: $k = 0.3$; $C = 0.3nd$; $C = \$1080$



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- Write the joint variation equation: $C = knd$.
- Substitute $C = 480$, $n = 2$, $d = 800$: $480 = k(2)(800) = 1600k$.
- Divide: $k = 480/1600 = 0.3$.
- Equation: $C = 0.3nd$.
- Find C when $n = 3$, $d = 1200$: $C = 0.3(3)(1200) = 0.3(3600) = \1080 .

9. Answer: $k = 1$; $w = xyz$; $w = 240$

- Write the equation: $w = kxyz$.
- Substitute $w = 210$, $x = 3$, $y = 5$, $z = 14$: $210 = k(3)(5)(14) = 210k \rightarrow k = 1$.
- Equation: $w = xyz$.
- Find w when $x = 6$, $y = 4$, $z = 10$: $w = (6)(4)(10) = 240$.

10. Answer: $k = \pi/3$; $V = (\pi/3)hr^2$; $V = 100\pi \text{ cm}^3$

- Write the joint variation equation: $V = khr^2$.
 - Substitute $V = 48\pi$, $h = 9$, $r = 4$: $48\pi = k(9)(16) = 144k$.
 - Solve for k : $k = 48\pi/144 = \pi/3$.
 - Equation: $V = (\pi/3)hr^2$.
 - Find V when $h = 12$, $r = 5$: $V = (\pi/3)(12)(25) = (\pi/3)(300) = 100\pi \text{ cm}^3$.
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